



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

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SECR1213-02 NETWORK COMMUNICATIONS

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GROUP NAME: Y-LINK

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1.0 NETWORK DEVICES

Devices	Description	Unit Price (RM)	Quantity	Total Amount (RM)
Router	Brand : TP - Link Model : Archer AX73 Ports : 5 Antennas : 6 Connectivity Type : Wireless Features : - Fully Featured Wi-Fi 6 - 1.5 GHz Triple-Core CPU - 1xGigabit WAN Port and 4x Gigabit LAN Ports - Network Security : • Real-Time IoT Protection • Malicious Site Blocker • DDoS Attack Prevention	495.00	6	2970.00
Switch	Brand : Netgear Model : GS752TPv2 Memory : 256 MB Power Budget : 760 W Features : - Has 48 Gigabit ports, 48 Power over Ethernet+ (PoE+) ports and 4 SFP (fiber) ports for 1 Gbps uplinks. Therefore, it is suitable to be used in each lab that has 30 workstations. - Supports up to 256 VLANs - Operating temperature for the switch is from 0 degree Celsius to 50 degree Celsius so it is optimal to be used in faculty and does not require a cooling mechanism. - Can host up to 512 Address	1260.00	13	16380.00

	Resolution Protocol (ARP) - Can support up to simple Layer 3 routing - Is Energy Efficient Ethernet (EEE), thus it reduces power consumption.			
Antivirus	Brand : Bitdefender Type : Software Package : Internet Security Features : - Real-time threat protection - Anti-ransomware - Firewall - Anti-fraud and safe browsing - Low system impact - Performance optimization - Centralized management (via GravityZone) - Web protection	120.00	65	7800.00
Cable	Brand : Cables Direct Online Type : Category 6 Bandwidth : Up to 550 MHz Data Transmission Speed : Up to 10 Gbps Features : - Supports 10Base-T (Ethernet), 100Base-T (Fast Ethernet) and 1000Base-T (Gigabit Ethernet) network environments. - Compatible with modems, computers, laptops, networking switches, routers, printers, etc. - Twisted-pair design to minimise electromagnetic interference - Supports PoE and PoE+ which enables power and data to be delivered over a single cable - Available in Shielded Twisted Pair (STP) or Unshielded Twisted	200 ft - 125.00 100 ft - 76.00	15 x 200ft 195 x 100ft	16695.00

	Pair (UTP) configurations.			
Wireless Access Points	Brand : Ruckus Wi-Fi Standard : Wi-Fi 6 (802.11ax) with dual radio (2.4 GHz and 5 GHz) Data Rate : 1148 Mbps for 2.4 GHz and 2400 Mbps for 5 GHz Ports : 1 Multigigabit Ethernet port and 1 Gbps Ethernet port Power : Supports 802.3bt and 802.3at PoE Features : - Has 4x4 Multiple-Input Multiple-Output (MIMO) Configuration - Supports Ruckus Beamflex which is an agile antenna array with multiple high-gain, directional antenna elements that are combined in real time to maximise range and coverage.	5230.00	6	31380.00
Modem	Brand : ARRIS SURFboard 33 DOCSIS 3.1 Multi-Gigabit Cable Modem - Compatible with DOCSIS 3.1 and 3.0. - Supports IPv4 and IPv6, internet browsing standards. - Provide 2 OFDM Downstream Channels and OFDMA Upstream Channels. - Features Dual Ethernet ports: 2.5 Gigabit Ethernet Port and 1 Gigabit Ethernet port, allowing separation in home networks with high-speed connections to multiple devices, ideal for maintaining stable and fast access for critical applications.	979.06	2	1958.12
Network Management	Brand : ManageEngine OpManager	1542.00	8	12336.00

Tool	Package : Professional Business Type : Software Features: - Real time monitoring tracks the availability, health and performance of network devices - Customizable dashboard provides visual insights into network status traffic and bandwidth usage with widgets and graphs - Proactive Alerts notifies administrators - NetFlow analysis monitors bandwidth usage and traffic patterns - Device configuration backup automatically backs up configuration of network devices			
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Budget Provided : RM2,000,000

Total price of required devices : RM89,519.12

Remaining budget : RM1,910,480.88

2.0 DISCUSSION

1. Are you surprised by the prices? How were you surprised?

Our total expenses for this project is estimated around RM89,519.12. We are not surprised by these prices because we expected such equipment to be more expensive. This is due to its advanced features and important function in guaranteeing a reliable and high-performance network infrastructure. Servers, routers, switches, antivirus software, cables, wireless access points and modems are all critical components for handling large-scale connectivity and security in a facility in our faculty building. As the estimated budget is still in our range of budget, we believe that the higher cost for networking devices is aligned with the complexity and durability that are necessary for such systems and we also believe these prices represent their long-term worth and reliability.

2. Have you ever considered cost as a factor for choosing networking devices?

Our main focus for the networking devices is on the quality, performance and the reliability of the components that are suitable for our faculty building project. Hence, we have not considered cost as a significant factor when choosing networking devices. We believe that focusing on cost often leads to performance issues or limited functionalities, which can result in higher expenses on the maintenance, replacements or upgrades.

3. What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Router :

For routers, the Netgear Nighthawk AX12 router has 6 ports compared to TP-Link Archer AX73's 5 ports, the Archer AX73 offers a more cost-effective solution for setups with moderate wired connections. TP-Link also provides 6 antennas, ensuring broad coverage, whereas Netgear's 8 antennas are optimized for extremely large spaces, which may be excessive for faculty use. The CPU in the TP-Link router balances performance and energy efficiency, whereas Netgear's 2.2 GHz Quad-Core CPU is ideal for high-demand environments but might consume more power. Advanced security is provided by both models, but TP-Link's Real-Time IoT Protection and DDoS Prevention features make it especially appropriate for protecting smart devices in a faculty building. In this situation, the TP-Link Archer AX73 is a better option for useful and secure wireless networking because of these qualities, as well as its affordable price and easy-to-use UI.

Feature	TP-Link Archer AX73	Netgear Nighthawk AX12
Ports	5 (1x WAN, 4x LAN)	6(1x WAN, 5x LAN)
Antennas	6	8
Processor Speed	1.5GHz Triple-Core CPU	2.2 GHz Quad-Core CPU
Security Features	Real-Time to IoT Protection, DDoS Prevention	Netgear Armor, WPA3 Security

Antivirus:

My group chose Bitdefender over other antivirus software like McAfee and Avira due to its superior balance of performance, security and centralised management. Real time threat protection and advanced threat defence in Bitdefender means protection for the lab PCs against both known and emerging cyber threats. One of the biggest advantages of Bitdefender is it was designed on a basis of low system impact, unlike McAfee which was previously known to be resource-intensive that slows down the system. Bitdefender ensures that resource-heavy apps, such as modelling tools and programming IDEs, work smoothly and without many issues. Furthermore, the centralised management of Bitdefender's GravityZone allows IT administrators to defend all lab PCs from a single console unlike Avira which lacks strong management tools. Additionally, Bitdefender offers advanced features related to ransomware remediation, network attack defence and device control, thereby making it perfect to ensure sensitive data security and unauthorised access prevention in shared lab environments. Because of its high performance and protection against almost everything, Bitdefender is more reliable and scalable than McAfee and Avira, especially in academic lab environments.

Feature	Bitdefender	Avira	McAfee
Real time threat protection	Advanced protection	Good	Strong
System performance impact	Low system impact	Lightweight but lacks advanced optimization for high performance tasks	Resource intensive

Centralized management	Yes	Limited centralised tools	Yes but less user-friendly
Network protection	Includes advanced firewall and network attack defence	Limited to basic protection	Good network protection with secure VPN

Modem:

We choose the ARRIS SURFboard 33 DOCSIS 3.1 Multi-Gigabit Cable Modem for our group project because it offers advanced features that align with the future-focused needs for our building. Unlike the LinkSys High Speed DOCSIS 3.0 24x8 AC1900, ARRIS SURFboard 33 DOCSIS 3.1 Multi-Gigabit Cable Modem allows download speeds of up to 5 Gbps. ARRIS S33 has features of Dual Ethernet ports, including 2.5 Gigabit Ethernet Port and 1 Gigabit Ethernet port, allowing separation in home networks with high-speed connections to multiple devices, ideal for maintaining stable and fast access for critical applications. Although the LinkSys High Speed DOCSIS 3.0 24x8 AC1900 simplifies the setup by combining modem and router functions, the ARRIS S33's performance features make it the best choice for our group project's goal of creating a better infrastructure in the faculty's building.

Feature	ARRIS SURFboard 33 DOCSIS 3.1 Multi-Gigabit Cable Modem	LinkSys High Speed DOCSIS 3.0 24x8 AC1900
DOCSIS standard	DOCSIS 3.1 with 2x2 OFDM/OFDMA channels (DOCSIS 3.0: 32x8 channels)	DOCSIS 3.0 with 24x8 channel bonding
Wi-Fi Capabilities	No Wi-Fi	Yes, Dual-Band Wi-Fi (AC1900)
Maximum Download Speed	Up to 5 Gbps with DOCSIS 3.1	Up to 960 Mbps with DOCSIS 3.0
Ethernet Ports	2 Ethernet Ports: 1x2.5 Gigabit Ethernet and 1x1 Gigabit Ethernet	4 Gigabit Ethernet Ports
Cost	Higher	More affordable

Network Management Tool:

Our group chose ManageEngine OpManager because of its good balance among affordability, scalability and feature comprehensiveness, which best fit our needs in a faculty lab environment. Unlike Auvik, which has very good automation and real-time visualisation at a high cost, OpManager gives robust monitoring tools and centralised management within budget limits. The intuitive interface that it has means that it greatly simplifies the work of network management without requiring advanced technical experience, making it perfect for university IT teams. Besides, compared to Datadog, which is designed for complex hybrid cloud environments, the focus of ManageEngine OpManager on on-premise network means it remains streamlined and cost-effective for faculty labs. Its easy scalability with the growth of the lab network only further makes it apt for managing devices and ensuring a secure, efficient learning environment.

Feature	ManageEngine OpManager	Datadog Network Monitoring	Auvik
Real time monitoring	Comprehensive, robust monitoring with alerts	Advanced real-time monitoring and analytics	Strong real time monitoring with cloud access
Scalability	Scales effectively without significant cost increases	Highly scalable for enterprise or hybrid setups	Scales easily but costly as devices increase
Cost	Affordable tiered pricing, suitable for limited budgets	Expensive for small networks	Higher subscription-based pricing per device
Automation	Moderate automation	Extensive	Advanced automation powered by AI/ML

3.0 MEETING MINUTES

Date/Time	15 Nov 2024, 7.10 p.m.		
Location	Online (Whatsapp Video Call)		
Agenda	1. Discuss which devices need to be in the new building		
	2. Divide devices to each group member		
Meeting Host	Harini Sangaran		
Attendance			
Name	Time	Reason For Absence	
Dayang Farah Farzana	7.10	-	
Farra Nurzahin	7.10	-	
Harini	7.10	-	
Safiya Nursyahadah	7.10	-	
Minutes			
No.	Item Discussed	Details	Person in charge
1.	Discussion on which devices is important	- All members suggested type of devices the new building should have	All members
2.	Room distribution	- Safiya is responsible for server room and video conference room - Farra is responsible for staff room and cisco lab - Dayang is responsible for two general purpose labs - Harini is responsible for embedded lab and hybrid classroom	All members
3.	Devices distribution	- Safiya is responsible for server and router - Farra is responsible for	All members

		modem and firewall - Dayang is responsible for antivirus, ethernet cable and network management tool - Harini is responsible for cable, switch and access point	
4.	Meeting ended	- At 9 p.m., Harini ends the meeting after all discussions have been done.	Harini

Date/Time	18 Nov 2024, 7.30 p.m.		
Location	Online (Google Meet)		
Agenda	1. Present each devices		
	2. Choose which brand is the best for each of the listed devices		
	3. Discuss question on Task 3		
Meeting Host	Farra Nurzahin		
Attendance			
Name	Time	Reason For Absence	
Dayang Farah Farzana	7.30	-	
Farra Nurzahin	7.30	-	
Harini	7.30	-	
Safiya Nursyahadah	7.30	-	
Minutes			
No.	Item Discussed	Details	Person in charge
1.	Discussion on which devices is the best	- Dayang presented her research of which antivirus and ethernet cable is the best - Farra presented her research of which modem and firewall is the best - Harini presented her research of which cable, switch and wireless access point is the best - Safiya presented her research of which server and router is the best	All members
2.	Decision making	- Dayang and Farra agree to combine firewall and	All members

		antivirus	
		- Harini takeover ethernet cable and combine it with cable - Each group member finalise decision on which brand is the best	
3.	Short discussion on question given on Task 3	- All members give comments each question given in Task 3	All members
4.	Meeting ended	- At 9.30 p.m., Farra ends the meeting after all discussions have been done.	Farra